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Dynamics of Networks

POLICY EVALUATION IN COLLABORATION NETWORKS

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Introduction

Public research funding has become one of the main **instruments used by governments to steer scientific production and innovation**. Beyond the direct financial support it provides, such funding is often expected to reshape the way researchers work together: by creating new partnerships, intensifying existing ones, and strengthening the collective capacity of research groups. Yet evaluating these policies is not straightforward. Scientific activity is inherently collaborative and embedded in networks, meaning that a policy targeted at some groups may affect not only the treated researchers, but also their collaborators, sometimes in ways that reinforce the policy's objectives and sometimes in ways that generate unintended consequences. In this sense, research funding is not just a "shock" to individual resources; it can also be a shock to the structure and dynamics of collaboration networks.

This project **evaluates the impact of a public research funding policy implemented at the group level.** Specifically, some research teams were selected in 2011 to receive support, and funding was rolled out starting in 2012. The core rationale behind such a policy is twofold: by funding selected groups, it may first improve the individual performance of their members (through greater resources, time, and capacity to launch projects), and second reshape collaboration dynamics by strengthening coordination and collective projects. In a network setting, these effects may also extend beyond direct beneficiaries: untreated researchers connected to treated members can be affected, either positively (knowledge diffusion, reputational effects) or negatively (reallocation of collaboration effort).

<u>To make these objectives concrete, we organize the analysis around the following research questions :</u>

- After 2012, do researchers in treated groups experience different changes in individual outcomes compared with researchers in non-treated groups ?
- Does the policy change collaboration activity, in terms of the probability of collaborating, the number of distinct partners, and the total collaboration volume ?
- Are these changes mainly driven by internal collaborations within the same group, or by external collaborations across groups ?
- Do we observe spillovers on untreated researchers: among non-treated individuals, do those who were connected to future treated researchers before 2012 evolve differently after the reform ?
- Does the policy affect link formation at the dyad level, and does the effect differ between treated–treated dyads and treated–untreated dyads, relative to untreated–untreated dyads ?
- Finally, do these effects vary by gender and by core versus non-core status ?

To answer these questions, we exploit the policy timing (pre-2012 vs post-2012) and compare treated to non-treated groups **using difference-in-differences methods with fixed effects**, complemented by event-study graphs to examine year-by-year dynamics and check pre-policy trends. The remainder of the report is organized as follows. Section I presents the data and descriptive statistics. Section II reports the estimated impacts on individual outcomes and collaboration measures, including spillovers and heterogeneity analyses. Section III studies link formation at the dyad level and provides robustness checks. Section IV concludes with a discussion and policy implications.

I - Descriptive analysis

This section provides a **descriptive reading of the data used in the study**. It aims to clarify the structure of the network (unit of observation, period covered, organization into groups) and to present the main characteristics of the individuals and groups observed, in order to anchor the interpretation of the empirical results in a precise understanding of the sample.

Our dataset comes from a collaboration file that has a naturally dyadic structure: each observation corresponds to a pair of individuals (i, j) observed in a year t, with information on the link, the intensity of the collaboration, and attributes of the individuals and their group. A key point, which strongly conditions the interpretation of descriptive statistics, is that the raw file does not necessarily list all possible pairs of individuals, but rather the observed dyads (depending on how the dataset was constructed). In other words, there is a risk of mechanical selection: if the dataset contains only "active" dyads (those that collaborated at least once over the window), then the statistics describe collaborations among people who already collaborate, not the probability of collaboration in the whole population. This implies that an increase (or decrease) in an aggregate indicator may come either from a real change in behavior, or from a change in the composition of observed dyads over time, which calls for a cautious reading of raw comparisons.

The analysis covers the **period from 2005 to 2019** and includes **9,031 individuals distributed across 255 groups.** For the overall description, we naturally distinguish two sub-periods, before 2012 and from 2012 onward, which correspond to the implementation of the policy in the data. The split of groups across categories is almost perfectly balanced: **128 treated groups versus 127 non-treated groups**. This symmetry is useful to describe the network without having a marginal category “artificially weigh” on the averages. *(appendix p.22)*

To better anchor the descriptive analysis, it is useful to describe the organization of the network by groups. **Most individuals are attached to a single group** (54.9% pre-treatment versus 53.8% post-treatment). The distribution is moreover very similar across the two periods: the share of individuals affiliated with two groups increases slightly (from 25.1% to 26.0%), the share affiliated with three groups remains practically unchanged (from 13.5% to 13.3%), and affiliations with four groups or more remain marginal. In other words, the affiliation structure observed in a given year is broadly stable before and after treatment. This does not prevent some mobility over the full period: some individuals change groups over the years, and it happens that some are observed successively in treated and non-treated groups in the full sample. *(appendix p.24)*

Before turning to individual characteristics, we can also describe how the overall network changes across periods. With the same population of 9,031 individuals, the **network becomes more connected** after 2012: the number of observed links rises from 29,005 to 52,500, implying a clear **increase in density** (0.00071 to 0.00129, about +80%). Clustering patterns are more nuanced. The average local clustering tends to overweight low-degree nodes (each node counts the same in the average), so we complement it with global clustering (transitivity), which captures closed triangles relative to all connected triplets and therefore emphasizes dense parts of the network, especially around highly connected researchers. In our data, **global clustering slightly increases** (0.259 to 0.268) **while average local clustering decreases** (0.457 to 0.440). The fact that global clustering remains well below average local clustering in both periods suggests that cohesion is still largely driven by a central core of strongly interconnected individuals. As discussed earlier, these remain raw descriptive comparisons and should be interpreted cautiously given the potential selection in which dyads are observed in the original file. *(appendix p.25)*

Based on this membership structure, we can now draw a more precise portrait of the individuals who make up the network, in order to better understand its composition and the main sources of heterogeneity. The profile of individuals highlights a heterogeneity that is classic in scientific collaboration networks. The network is composed of **57.7% men and 42.3% women**, and it is **mainly made up of non-core members (76%) rather than core members (24%)**. Age, reconstructed with reference to 2005, is essentially **concentrated between 25 and 45 years old**, with a peak around the thirties/forties and few individuals at extreme ages. Finally, disciplinary affiliation, non-exclusive since researchers may belong to several fields, is **highly concentrated: field_3 concerns**

all individuals, while field_1 and field_2 cover most of the sample (around 80%), with fields 4 to 8 remaining marginal. Overall, these statistics describe a rather male network, dominated by non-core profiles, and composed of a population of intermediate age at the beginning of the period. *(appendix p.22 to 30)*

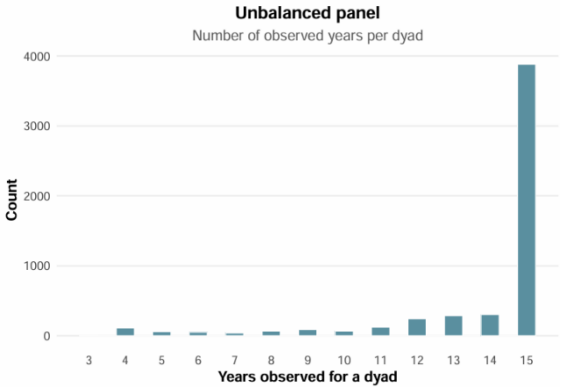
As, in what follows, we implement a **Difference-in-Difference framework on the sample that satisfies the selected selection criteria**, it is useful beforehand to examine its main characteristics in order to clarify what the comparison is based on and how to interpret the results. A central point is the **stability of group membership over time**: individuals remain attached to the same group and **do not move from a non-treated group to a treated group** (nor the reverse). This stability is particularly important in a Difference-in-Difference, because it guarantees a clear separation between treated and control units and avoids individuals changing status over the period, which would blur the comparison. *(appendix p.33)*

	Treated		Non Treated		Difference t-test	
	Mean	Sd	Mean	Sd	b	p
Share of females	0.394	0.489	0.434	0.497	0.10	(0.27)
Share of core members	0.273	0.446	0.140	0.348	-0.49	(0.00)
Age	40.492	9.359	43.001	9.932	0.06	(0.00)
Outcome 1	2.400	3.107	3.521	5.271	0.47	(0.00)
Outcome 2	23.013	49.859	40.304	77.506	0.75	(0.00)
Outcome 3	2.564	5.077	3.721	6.431	0.45	(0.01)
Share in field 1	0.822	0.383	0.955	0.208	0.16	(0.00)
Share in field 2	0.812	0.391	0.955	0.208	0.18	(0.00)
Share in field 3	1.000	0.000	1.000	0.000	0.00	-
Share in field 4	0.071	0.258	0.145	0.353	1.03	(0.00)
Share in field 5	0.071	0.258	0.090	0.288	0.27	(0.36)
Share in field 6	0.041	0.199	0.100	0.300	1.41	(0.01)
Share in field 7	0.015	0.124	0.014	0.116	-0.12	(0.82)
Share in field 8	0.063	0.243	0.027	0.163	-0.57	(0.01)
Share with any collaboration	0.515	0.343	0.563	0.355	0.09	(0.06)
Number of partners	1.736	2.018	2.759	4.233	0.59	(0.00)
Total collaboration volume	2.572	3.698	4.640	8.151	0.80	(0.00)
Number of internal partners	0.996	1.246	1.374	1.912	0.38	(0.01)
Number of external partners	0.740	1.198	1.384	2.816	0.87	(0.00)
Internal collaboration volume	1.589	2.558	2.760	5.515	0.74	(0.00)
External collaboration volume	0.983	2.039	1.879	3.798	0.91	(0.00)
Observations	1162		221		1383	

Moreover, even if statuses remain stable over time, it is useful to assess to what extent the two sets resemble each other before treatment, in order to position the starting point of the comparison. Overall, some dimensions appear close: the **gender composition is very similar**, which suggests satisfactory comparability on this aspect despite a slight gap. By contrast, more marked differences emerge on other characteristics: **treated units include a higher share of core members while control units are, on average, older**. These **gaps are also reflected in average performance levels** (outcomes 1 to 3) and, even more so, in collaborative activity: **control mainly stand out through a higher number of partners and larger collaboration volumes**. This difference concerns both internal and external collaborations, **with particularly pronounced gaps on the external dimension**.

In sum, these elements indicate that the treated/control comparison rests on well-defined and stable statuses, which is favorable for the interpretation of a diff-in-diff. By contrast, the level gaps observed before treatment reinforce the idea that the analysis should not be read as a simple comparison of levels between groups, but rather as a comparison of changes over time, keeping in mind that the two sets do not start from exactly the same point.

Furthermore, we consider a descriptive analysis of the dyadic database to be essential in order to better understand the network dynamics and the mechanisms underlying link formation in our difference-in-differences (DiD) framework. Since the baseline levels of treated and control groups differ, it is important to carefully examine the structure of the data prior to estimation. It is important to clarify that this group-level analysis is not conducted on the full dyadic dataset. Instead, it relies on **a restricted subsample** defined by **selection criteria 1 and 2**. This restriction ensures that individuals included in the comparison satisfy the same baseline eligibility conditions, thereby providing a **common support** between treated and untreated units. This design allows us to compare structurally comparable groups and reduces concerns that observed differences simply reflect compositional heterogeneity. In particular, the below histogram analysis shows that most dyads are observed both before and after 2012.



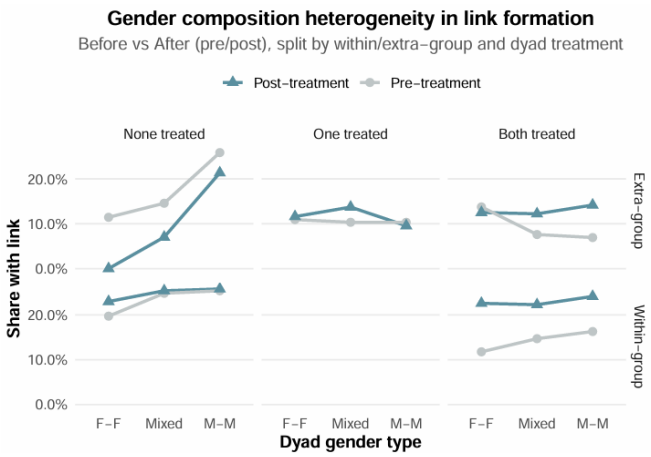
Therefore, the DiD estimates primarily rely on within-dyad variation rather than on compositional changes in the network. This substantially reduces concerns related to dynamic selection bias and strengthens the credibility of our identification strategy, especially given that most dyads are observed over a long-time horizon (up to 15 years).

Our identification strategy is based on a difference-in-differences framework including dyad and year fixed effects. We exploit both temporal variation and differences in treatment exposure while controlling for unobserved time-invariant heterogeneity. **Under the parallel trends assumption**, the estimated coefficient can thus be interpreted as the causal effect of the treatment on collaboration formation.

Before proceeding to the causal analysis, it is informative to examine the aggregate evolution of collaboration patterns over time. Figures 1 and 2 provide complementary perspectives on network dynamics at the dyad level. The **Figure** *(appendix p.40)*, displays the average collaboration intensity per dyad, measured as the mean number of collaborations per pair each year. A clear upward trend emerges over the full period. Collaboration intensity increases steadily from 2005 onward, with a moderate slowdown around 2010–2011, followed by a pronounced acceleration after 2012. The post-2012 period is characterized by a sustained rise and eventual stabilization at a higher level. This pattern suggests that collaboration deepening; conditional on existing dyads; becomes stronger over time. The **Figure** *(appendix p.39)*, in turn, reports the probability that a dyad form at least one collaboration link in a given year (The dataset includes only **"active" dyads**, i.e., dyads that are observed in the panel.). The share of dyads with an active link increase substantially between 2005 and 2014, indicating a progressive densification of the network. After 2014, the probability stabilizes at a relatively high level, with some fluctuations but no evidence of a structural reversal in the overall upward trend. Taken together, these figures reveal two important features of the data. First, the network exhibits a **long-run expansion** in both the extensive margin (probability of link formation) and the intensive margin (collaboration intensity). Second, the upward trajectory begins well before 2012, suggesting the presence of **secular trend**s in collaboration that must be carefully accounted for in the econometric design. These descriptive patterns reinforce the relevance of our difference-in-differences framework with dyad and year fixed effects**. Year fixed effects** absorb the common upward trend in collaboration intensity visible in both figures, while **dyad fixed effects** control for persistent pair-specific heterogeneity. Consequently, identification does not rely on aggregate growth in collaboration, but rather on differential changes in treated versus control dyads relative to these common trends.

Building on the aggregate trends described above, we examine whether post-2012 dynamics differ by treatment exposure**. The Figure** *(appendix p.42)* shows that the probability of link formation increases markedly after 2012 for dyads in which both members are treated, while non-treated dyads largely maintain their pre-treatment levels. The magnitude of the increase for both-treated dyads is economically meaningful, pointing to a direct and immediate effect of the policy. Moreover, partially treated dyads exhibit a moderate upward shift relative to fully untreated pairs, suggesting the presence of spillover effects through indirect exposure to treated individuals. Overall, these heterogeneous dynamics support the interpretation of a strong direct treatment effect. Further insight emerges **when distinguishing between extra-group and within-group dyads** *(appendix p.44)*. The figure reports link probabilities before and after the policy **by dyadic treatment status**. For **extra-group dyads**, pre-treatment link probabilities decline markedly as we move from none-treated to both-treated pairs, indicating that cross-group collaborations involving treated units were already less frequent prior to the reform. After 2012, probabilities become more compressed across treatment categories, driven mainly by a decline among untreated dyads. In contrast, exposure to treatment does not generate a substantial increase in cross-group ties. This pattern suggests a relative weakening of external collaborations, particularly among untreated pairs. **Within-group dyads** display a different dynamic. Collaboration probabilities are substantially higher overall, reflecting stronger internal cohesion. While pre-treatment levels are already elevated across categories, the post-treatment period is characterized by a pronounced increase for fully treated dyads. This indicates that the policy primarily **reinforced collaborations within treated groups** rather than stimulating external connections. Taken together, these descriptive patterns point to a reallocation of collaboration activity following the reform: **external** links **weaken**; especially among untreated dyads; whereas internal collaborations **within** treated groups **intensify**. Such dynamics are consistent with a consolidation mechanism, whereby policy exposure strengthens within-group integration more than cross-group openness.

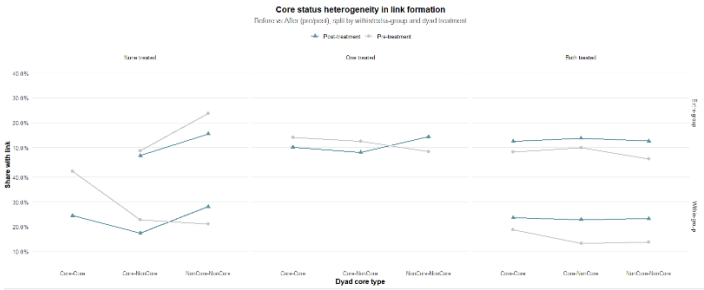
The figure below further refines the analysis by introducing **gender composition into the within- versus extra-group comparison.**



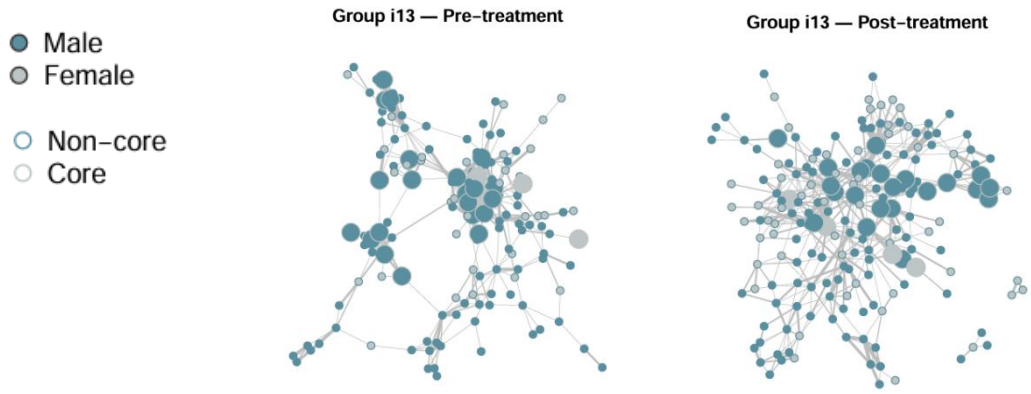
The treatment effect is primarily driven by dyads in which both members are treated. In this case, post-treatment **extra-group** link probabilities increase, particularly for **Mixed** and **M--M** dyads, whereas **F--F** dyads remain comparatively stable. This pattern suggests that the policy stimulates outward-oriented collaboration among treated units, **encouraging cross-group expansion** rather than merely reinforcing existing internal ties. **Within groups**, however, the changes are comparatively modest across gender categories.

While collaboration probabilities remain broadly stable, a generalized upward adjustment is observed across all gender compositions. An additional striking feature concerns **untreated dyads** in the **extra-group** dimension: for none-treated pairs, collaboration probabilities decline across all gender compositions in the post-treatment period. In contrast, **within-group** probabilities for untreated dyads remain broadly stable. This asymmetry reinforces the interpretation of a **reallocation mechanism**; whereby **treated dyads** expand their **external collaborations** while **untreated dyads** experience a relative **contraction**. Overall, the effect is heterogeneous across gender compositions and is largely driven by treated–treated dyads extending their external connections. The particularly strong response observed among **Mixed dyads** also points to an indirect or diffusion effect, suggesting that treatment exposure may **reshape collaboration patterns beyond purely homogeneous pairs.**

The figure below introduces heterogeneity by **core status** and further clarifies the mechanism underlying the treatment effect.

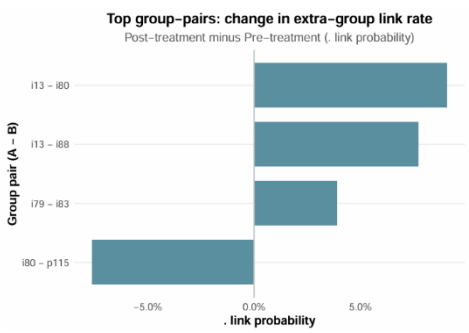


The most economically meaningful adjustments occur when both members of the dyad belong to treated groups. In this configuration, we observe a clear **increase in extra-group link formation**, particularly among **NonCore–NonCore** dyads, pointing to a strong outward-oriented expansion of the network. Within groups, the post-treatment period is characterized by a noticeable rise in **Core–NonCore** and **NonCore–NonCore** collaborations, suggesting that treated core members become more central in internal link formation. Rather than merely reinforcing pre-existing core–core cohesion, the policy appears to **activate cores as connectors toward peripheral actors**. In contrast, partial exposure **(one treated)** generates only limited and heterogeneous adjustments, providing **weak evidence of spillover-driven** restructuring. Dyads with **no treated** members largely reflect **secular trend**s, with no structural break attributable to the policy. Overall, these patterns indicate that the treatment **strengthens the strategic role of core members** as central drivers of link formation, both by **expanding cross-group connections** and by **integrating non-core actors internally**. The reform therefore operates less through simple densification of peripheral ties and more through a reorganization of network centrality around treated core actors. As illustrated by the case of group **"i13,"** the post-treatment network appears substantially denser than in the pre-treatment period, with non-core members exhibiting stronger internal integration.

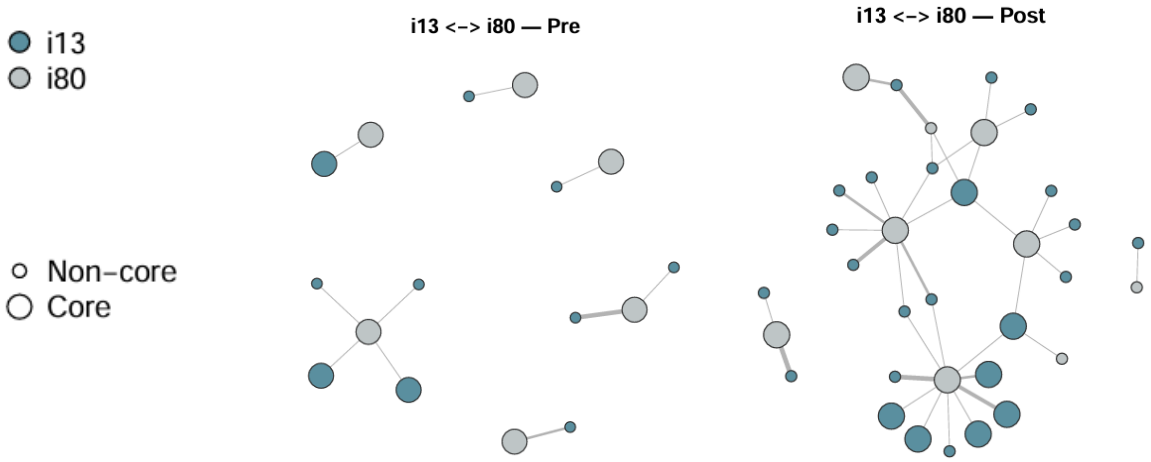


Within the restricted subsample, we investigate the inter-group dimension of the reform by identifying the **between-group pairs** exhibiting the largest change in extra-group link probability (post minus pre).

Among these, the **pair i13–i80** stands out as displaying one of the most pronounced increases after 2012. While such leading pairs may reflect scientific complementarity, geographical proximity, institutional alignment, or heightened responsiveness to the reform, these figures represent raw variations and do not, at this stage, constitute formal causal identification. To better understand the underlying mechanism, we therefore examine the pre- and post-treatment network structures of i13 and i80 in greater detail. In the **pre-period**, cross-group collaboration is



sparse and fragmented, with **small isolated** components and **no dominant** bridging actors; core members are present but do not occupy central inter-group positions. In contrast, the **post-treatment** network becomes markedly **denser** and more **integrated**: a large connected component emerges, and core actors move into structurally central positions, forming multiple cross-group ties and effectively serving as **brokers** between the two communities. Complementary **network statistics** *(appendix p.131)* confirm a substantial expansion and intensification of activity in both groups, reflected in strong increases in the number of active nodes, links, average degree, and aggregate collaboration weight. However, the structural adjustments are **heterogeneous**. In some instances, integration rises alongside increased centralization, while in others expansion is accompanied by diffusion toward non-core members and mild fragmentation. Importantly, although core actors experience significant absolute growth in collaboration volume, their relative dominance does not systematically increase and may even decline, indicating **partial redistribution of collaborative intensity toward peripheral members**. Gender dynamics also evolve, with both male and female connectivity rising, albeit with varying relative gains across groups. Taken together, these findings suggest that the **treatment** does not merely raise collaboration mechanically; rather, **it reorganizes network architecture by expanding participation, reshaping concentration patterns, and redefining the distribution of collaborative influence.**



II - Impact of treatment on individual outcomes

This section **evaluates the effect of the policy on individual outcomes and on the structure of collaborations**. The challenge is twofold: to **measure an average effect on the trajectories of individuals exposed to the treatmen**t, and to understand through which network mechanisms this effect can materialize. Indeed, in the spirit of the course, a policy targeted at groups can modify not only activity levels, but also the architecture of the network (internal densification, reconfiguration of links, substitutions between internal and external collaborations), and **can potentially generate externalities on non-treated individuals** through collaboration links.

The analysis relies on a Difference-in-Differences (DiD) framework, which compares the evolution of outcomes for individuals affiliated with treated groups to that of similar individuals affiliated with non-treated groups, before and after the implementation of the policy. Since the treatment occurred in 2012, we define pre-treatment: $t < 2012$ and post-treatment: $t \geq 2012$.

In the DiD logic, identification comes from a **comparison of evolutions** (and not levels). This is particularly relevant here because, as we saw in the descriptive section, treated and control groups may differ in levels before treatment. **DiD precisely makes it possible to net out permanent differences between individuals and common shocks, in order to interpret the effect as a divergence in trajectories associated with the policy**, under the parallel-trends assumption: in the absence of treatment, treated and control groups would have evolved similarly.

In this framework, we estimate the reference DiD specification:

$$y_{i,g,t} = \alpha + \beta T_{g_i} \times Post_t + \gamma X_{i,t} + \delta_t + \theta_i + \varepsilon_{i,g,t}$$

where $y_{i,g_i,t}$ is an outcome variable, T_{g_i} is the treatment variable (1 if treated and 0 otherwise, and is specific to the group if i, g), $Post_t$ is the post treatment period, $X_{i,t}$ is a vector of control variables, θ_i is the individual fixed effect, δ_t is a year fixed effect, and $\varepsilon_{i,g,t}$ is the error term.

To implement this identification strategy, we **first need an individual-level panel consistent with the DiD equation**. However, the raw data describe a network in dyadic form (pairs (i, j) observed in a year t) and these dyads are bilateral: if (i, j) is present, (j, i) is not. We therefore transform the dataset by duplicating observations while switching identities, and then aggregating in order to obtain an individual (i) x group (g) x year (t) panel.

In the empirical implementation, we estimate these models with the **fixest** package, which makes it possible to efficiently include high-dimensional fixed effects. The baseline specification is written as:

$$\text{feols(y} \sim \text{group_treated:post} \mid \text{id + year, vcov = } \sim \text{group_id)}$$

where the interaction "group_treated:post" corresponds to the term $T_{g_i} \times Post_t$. **Individual fixed effects absorb all time-invariant heterogeneity** (for example, initial performance level, discipline, persistent position in the network), while **year fixed effects capture general trends and common shocks affecting the entire network**. **Standard errors are clustered at the group level**, because treatment assignment is done at that level and outcomes may be correlated among individuals within the same group. The interpretation of the coefficient $\boldsymbol{\beta}$ is therefore as a difference in changes: it measures the average change after treatment for the treated relative to the change observed for the controls.

II.1 - Direct effects: Average impact on members of treated groups

a) Average effect on individual outcomes

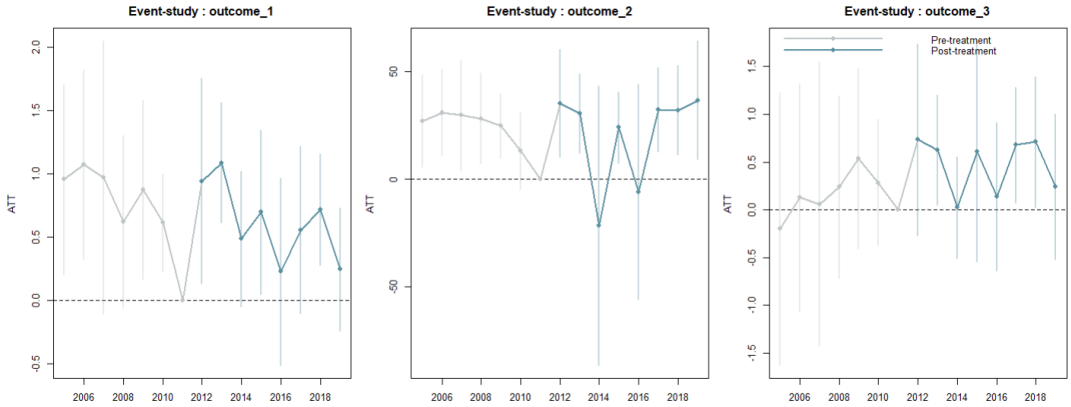
We first estimate the effect of the treatment on the three yearly individual outcomes (outcome_1, outcome_2, and outcome_3) using the baseline DiD specification. In this approach, the coefficient associated with $treated \times post$ measures the average difference in the post-treatment change between treated and control units, once fixed heterogeneity across individuals and common year shocks have been taken into account.

Dependent variable	outcome_1	outcome_2	outcome_3
Treatment x Post	-0.0952 (0.1544)	-1.434 (10.97)	0.3159 (0.2661)
Observations	20,553	20,553	20,553
R^2	0.64925	0.31933	0.44503
Within R^2	4.1e-5	8.2e-6	0.00015

Notes: *Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1*

The results do not provide evidence of a statistically significant average effect on the individual outcomes: the coefficients are small in magnitude and estimated with limited precision. In other words, at this stage, we do not observe a clear average divergence in the trajectories of the individual outcomes attributable to the treatment. *(appendix p.67)*

This result should nevertheless be interpreted with caution for two reasons. First, an average effect may mask effects that unfold gradually: collaborations, the diffusion of information, or the ramp-up of projects can take time before being reflected in individual indicators. Second, the credibility of the DiD relies mainly on the behavior of trajectories before treatment, since the key assumption is that of parallel trends. This is why we immediately complement the analysis with an event-study, which estimates a relative effect for each year (relative to a pre-treatment reference year). This specification provides two pieces of information. First, it shows whether the post-treatment effect is mainly concentrated in the first years, whether it increases over time, or whether it fluctuates. Second, it allows us to assess whether the pre-treatment coefficients are overall close to zero; if not, this signals pre-existing dynamics between treated and control units that should be kept in mind.



The event-study profiles indicate that, for outcome_1 and outcome_2, the pre-treatment coefficients are not systematically centered around zero, which suggests the existence of differential dynamics between treated and control units already present before 2012. In this context, the parallel-trends assumption is weakened for these two variables: the diff-in-diff estimates may partly capture the continuation of pre-existing trends rather than a causal effect of the policy. The post-treatment coefficients should therefore be interpreted with caution and as descriptive indications of a divergence in trajectories rather than as a measure entirely attributable to the treatment. By contrast, outcome_3 shows a pre-treatment profile that is more centered around zero, which makes identification more credible for this variable, even though the post-treatment effects remain overall modest and often imprecisely estimated. *(appendix p.77)*

In sum, no clear average effect emerges for the individual outcomes. The pre-existing dynamics observed for outcome_1 and outcome_2 weaken the causal interpretation of the DiD for these variables, while outcome_3 offers more credible identification but without a clear post-treatment effect. We therefore turn to the collaboration outcomes to analyze the network mechanisms more directly.

b) Average effect on collaboration outcomes

We then apply the same DiD specification to the collaboration variables in order to identify through which channels the policy modifies how the network operates. The selected indicators allow us to distinguish: an extensive margin, the fact of having at least one collaboration ($link_ind$); an intensive margin, the number of partners ($nb_partners$) and the total volume of collaborations ($total_collab$) and we decompose these measures into collaborations that are internal to the group ($n_internal_partners$ and $internal_collab$) and external ($n_external_partners$ and $external_collab$), which makes it possible to assess whether the policy mainly densifies interactions within groups or whether it also promotes openness toward the outside.

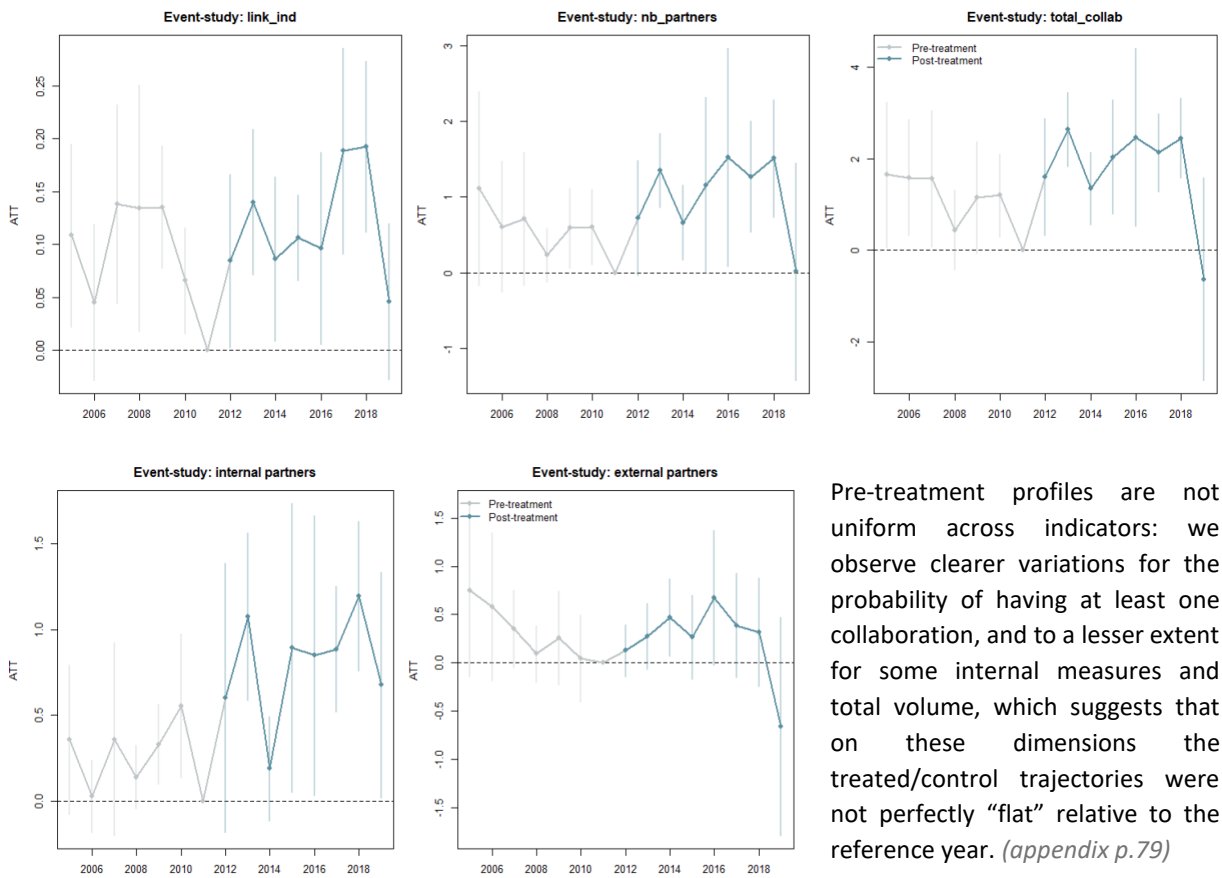
Dependent variable	link_ind	nb_partners	total_collab	n_internal_ partners	n_external_ partners	internal_ collab	external_ collab
Treatment x Post	0.0286. (0.0157)	0.4944* (0.1886)	0.7080* (0.3278)	0.5404** (0.1573)	-0.0460 (0.0884)	1.139*** (0.2937)	-0.4310** (0.1477)
Observations	20,553	20,553	20,553	20,553	20,553	20,553	20,553
R^2	0.34858	0.51941	0.54755	0.47560	0.46699	0.56047	0.44118
Within R^2	0.00017	0.00117	0.00077	0.00413	0.0000194	0.00506	0.00059

Notes: *All specifications include individual and year fixed effects. Standard errors are clustered at the group level (group_id). Significance levels: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1.*

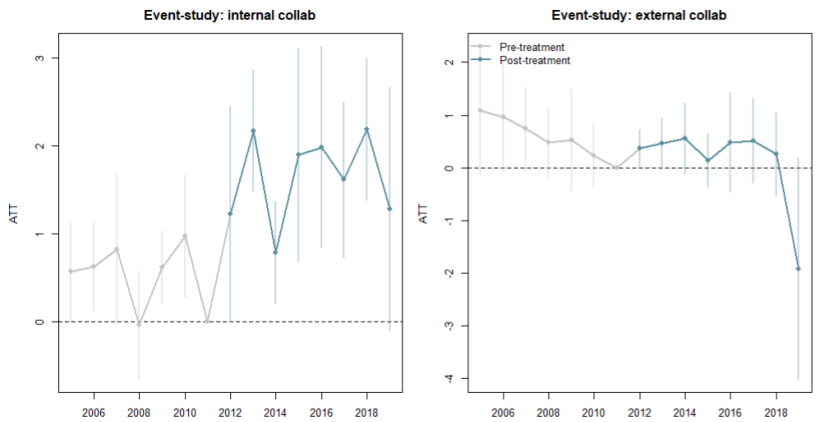
Using a 5% significance threshold, the effect on the extensive margin is limited: the increase in the probability of having at least one collaboration is not statistically significant. By contrast, the results are clearer on the intensive margin: the number of partners increases significantly (+0.494) and the total volume of collaborations also increases (+0.708). In other words, the policy is associated with an intensification of collaborative activity rather than with an increase in the mere fact of collaborating. *(appendix p.69)*

The internal/external decomposition highlights a central result. Internal collaborations increase strongly: the number of internal partners rises by +0.540 and internal collaboration volume by +1.139, both statistically significant effects. By contrast, the external dimension does not expand: the number of external partners does not change significantly, while external collaboration volume decreases significantly (−0.431). Taken together, these results suggest within-group densification accompanied by partial substitution away from the external margin: activity increases mainly within treated groups, without a robust expansion of collaborations outside the group, and with a decline in external volume.

To complement this average-effect interpretation, we estimate an event-study on the collaboration outcomes in order to track the evolution of effects year by year and to assess the possible presence of pre-treatment dynamics.



Pre-treatment profiles are not uniform across indicators: we observe clearer variations for the probability of having at least one collaboration, and to a lesser extent for some internal measures and total volume, which suggests that on these dimensions the treated/control trajectories were not perfectly "flat" relative to the reference year. *(appendix p.79)*



By contrast, the total number of partners and the external indicators appear more stable before treatment, which makes the reading of post-treatment gaps more credible.

After treatment, the coefficients are broadly in the same direction as the DiD, with effects that are most often positive on the overall and internal measures, while the external indicators remain less regular and more sensitive to noise toward the end of the period.

The decomposition confirms that this dynamic is mainly driven by the internal dimension. The coefficients associated with internal partners and internal volume are more consistently positive and larger in magnitude after treatment, which reinforces the idea of within-group densification.

By contrast, the profiles for the external dimension are more unstable: the coefficients are often smaller and frequently compatible with zero, and some years at the end of the period display substantial uncertainty (very wide confidence intervals). These extreme points should therefore be interpreted with caution and should not drive the conclusion on their own.

II.2 - Indirect effects: Average impact on untreated pre-collaborators of treated members

Beyond the **direct effects on members of treated groups, it is important to examine potential indirect effects on individuals whose group is not treated**. In a network context, the treatment of a group can indeed affect untreated individuals through their collaboration ties with treated members: depending on the mechanisms at play, these effects may be positive (knowledge spillovers, complementarities, reputational effects) or negative (crowding-out, reallocation of collaborations, concentration of activity within treated groups). The analysis therefore consists in **comparing, among untreated individuals, those who were exposed before 2012 to treated collaborators with those who were not**.

Concretely, exposure is constructed from pre-treatment, out-of-group collaborations. Two complementary measures are used: *(i)* an indicator equal to 1 if an untreated individual had at least one treated collaborator before the reform and *(ii)* an intensity measure equal to the number of distinct treated collaborators observed before the reform. We then estimate, in the sample of untreated individuals only, a DiD-type specification where the effect of interest is captured by the interaction **Post x Exposure**:

$$y_{i,t} = \alpha + \beta Exposure_i \times Post_t + \gamma X_{i,t} + \delta_t + \theta_i + \varepsilon_{i,t}$$

Where $Exposure_i$ measures individual i's pre-treatment exposure to treated partners (either binary or intensity). The coefficient β therefore indicates whether, after 2012, exposed untreated individuals evolve differently from non-exposed untreated individuals, all else equal. The estimates include individual and year fixed effects, with standard errors clustered at the group level. *(appendix p.82)*

Dependent variable	outcome_1	outcome_2	outcome_3	link_ind	nb_partners	total_collab
Post × Exposed	-0.0225 (0.2443)	1.750 (17.24)	-1.644** (0.4577)	-0.1853*** (0.0359)	-0.6894. (0.3505)	-1.050 (0.6007)
Observations	3,247	3,247	3,247	3,247	3,247	3,247
R^2	0.75110	0.25214	0.52374	0.39434	0.63383	0.62909
Within R^2	2.9e-6	4.26e-6	0.00570	0.01366	0.00277	0.00170

Dependent variable	n_internal_partners	n_external_partners	internal_collab	external_collab
Post × Exposed	-0.6871* (0.2953)	-0.0022 (0.1631)	-1.546* (0.4612)	0.4958 (0.2828)
Observations	3,247	3,247	3,247	3,247
R^2	0.54309	0.59508	0.65667	0.52008
Within R^2	0.01270	4.27e-8	0.01309	0.00061

Dependent variable	outcome_1	outcome_2	outcome_3	link_ind	nb_partners	total_collab
Post × #treated partners (pre)	-0.0477 (0.0269)	-0.7061 (1.168)	-0.2226*** (0.0356)	-0.0089* (0.0030)	-0.0653** (0.0158)	-0.0693 (0.0526))
Observations	3,247	3,247	3,247	3,247	3,247	3,247
R^2	0.75170	0.25223	0.53024	0.38952	0.63450	0.62897
Within R^2	0.00241	0.00013	0.01927	0.00580	0.00459	0.00137

Dependent variable	n_internal_partners	n_external_partners	internal_collab	external_collab
Post × #treated partners (pre)	-0.0705** (0.0192)	0.0052 (0.0304)	-0.1568*** (0.0143)	0.0876 (0.0629)
Observations	3,247	3,247	3,247	3,247
R^2	0.54864	0.59510	0.66077	0.52146
Within R^2	0.02469	4.29e-5	0.02487	0.00349

Notes: *Untreated sample only. Individual and year fixed effects included; standard errors clustered by group_id. Post x Exposed compares the post-2012 change in outcomes for untreated individuals exposed to treated collaborators before 2012 versus untreated individuals not exposed. Post x #treated partners (pre) is the same idea with a continuous exposure: β measures how the post-2012 change varies per additional treated collaborator observed before 2012. Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1.*

The estimates overall suggest the **presence of negative spillovers on untreated individuals who were exposed to the treated network before the reform**. Using the binary exposure measure, no statistically significant effect is found for outcome_1 and outcome_2, while outcome_3 decreases significantly (−1.644). For collaboration indicators, exposure is associated with a marked contraction in activity: the probability of having at least one collaboration drops sharply (−0.185), and the effects are concentrated on the internal dimension, with a decline in the number of internal partners (−0.687) and in the volume of internal collaborations (−1.546), both statistically significant. By contrast, external collaboration variables show no significant change, indicating that adjustment occurs mainly "within" rather than through a reallocation toward external ties.

The intensity-based exposure measure confirms and strengthens this diagnosis. Each **additional treated collaborator observed before 2012 is associated with a larger decline in outcome_3** (−0.223) and **with a gradual reduction in collaborative activity**: a decrease in the probability of collaborating, in the total number of partners, in the number of internal partners, and in internal collaboration volume, with effects that are statistically significant at the 5% level. Again, coefficients related to external collaboration remain small and statistically insignificant.

Taken together, these results are more consistent with **negative externalities (crowding-out or link reallocation)** than with positive diffusion. The intuition is straightforward: after 2012, treated researchers appear to concentrate their collaborations more within their own group. Exposed untreated individuals (those who collaborated with treated researchers before the reform) therefore lose part of these "bridging" collaborations. Because projects are carried out in pairs, this loss is not automatically offset by new external partners. As a result, the collaborative activity of exposed individuals contracts, which shows up mainly as a decline in collaborations within their own group (fewer internal partners and lower internal volume), while the external dimension does

not display robust changes. In other words, the network does transmit the policy's effect beyond the directly treated units, but rather in the form of an unfavorable adjustment for some untreated individuals who were connected to treated researchers before 2012.

II.3 - Heterogeneity of effects

After estimating the average effect of the program on individual outcomes and on collaborations, it is useful to **check whether this effect is uniform for all individuals**. In practice, the same policy may benefit some profiles more than others, and it **can also produce indirect effects through pre-existing relationships**. This is why we complement the analysis with a heterogeneity study, distinguishing, on the one hand, men and women, and on the other hand, core and non-core members within groups. *(appendix p.86)*

a) Heterogeneity by gender

For men, the direct effect significant at the 5% level mainly concerns the structure of collaborations: the program **increases the number of internal partners** (+0.566) **and the volume of internal collaborations** (+1.314), while **reducing the volume of external collaborations** (−0.910). In other words, the impact translates into a strengthening of internal collaboration, accompanied by a contraction of external collaboration. By contrast, no robust effect at the 5% level appears on individual outcomes.

For women, we observe a very similar pattern on the internal dimension: the program significantly **increases the number of internal partners** (+0.474) **as well as the volume of internal collaborations** (+0.869). However, unlike for men, the effect on external collaborations is not established at the 5% threshold, and individual outcomes do not show a robust effect either. *(appendix p.82)*

These results therefore suggest that, for direct beneficiaries, the program first acts as a lever to densify internal collaborations, in a fairly comparable way between men and women. The next question is whether this refocusing toward internal collaboration is accompanied by effects on the surroundings of treated groups: do untreated individuals, but initially connected to future treated individuals, benefit indirectly from the policy, or do they instead experience a displacement of opportunities? This is precisely the purpose of the spillover analysis. *(appendix p.89)*

When exposure is defined in a simple way (having had at least one link before the reform), we observe a clearly negative signal that is similar by gender: after 2012, the **probability of collaborating decreases for men** (−0.181) **as well as for women** (−0.190), **and the volume of internal collaborations declines in both groups** (around −1.55). **For men, we additionally observe a significant decrease in an individual outcome** (−2.26), suggesting that exposure can also affect certain dimensions of performance.

When exposure is measured by intensity (the more partners a person had who will become treated, the more exposed they are), **differences between men and women become clearer**. For women, exposure intensity is associated with a broader deterioration: a decrease in several individual outcomes and a marked contraction in collaboration activity, with a decline in the probability of collaborating (−0.022), the total number of partners (−0.329), and the total volume (−0.486), and a particularly visible decrease on the external side (fewer external partners and lower external volume). For men, intensive exposure also reduces the probability of collaborating (−0.0039) and internal activity (fewer internal partners and lower internal volume), but it is simultaneously accompanied by an increase in external collaborations, which suggests substitution: exposed untreated men seem to offset the internal decline by a relative refocusing toward the outside.

In sum, on the side of direct beneficiaries, the program has a robust and fairly similar effect for both sexes: it strengthens internal collaborations. The main difference is that, for men, this strengthening is also

accompanied by a significant decrease in external collaborations, whereas this effect is not established for women. On the spillover side, exposure is overall unfavorable in both groups, but it manifests differently: for women, higher exposure seems to lead to a general contraction (collaborations and individual outcomes), while for men it looks more like a recomposition of activity (less internal, but more external).

b) Heterogeneity by core

In light of these gender-based results, it is relevant to examine a second source of heterogeneity directly related to the network structure: **core versus non-core status**. In a collaboration network, central actors (core) generally have more connections and visibility, which may allow them to capture the opportunities created by the program more easily. Conversely, peripheral actors (non-core) may benefit more modestly, or even be more exposed to displacement effects. We therefore repeat the same approach by distinguishing core and non-core members, first analyzing the direct effect on beneficiaries, and then spillovers on exposed untreated individuals.

For core members, the program's direct effect is clearly positive and significant at the 5% level: we observe an **increase in an individual outcome** (+1.797) and, above all, a marked rise in collaboration activity. The **total number of partners increases** (+2.396), **the total volume of collaborations rises** (+3.455), **and this dynamic is particularly driven by internal collaboration, with more internal partners** (+1.783) **and higher internal volume** (+2.938). In other words, the gains are relatively broad and confirm that central actors appear to be those who benefit the most from the program, mainly through an intensification of collaborations.

For non-core members, the direct effect is more targeted: **the program translates into a strengthening of internal collaboration** (more internal partners: +0.315; higher internal volume: +0.802), **but also into a decrease in external volume** (−0.573). Individual outcomes do not show a robust effect at the 5% level, suggesting that for peripheral actors the main consequence of the program is a reorientation of collaborations toward the inside rather than a clearly measurable individual gain. *(appendix p.95)*

On the spillover side, exposure defined as "at least one link" before the reform is associated with a significant decrease in an individual outcome 3 for core members (−2.960) as well as for non-core members (−1.162). In terms of collaborations, exposed non-core individuals experience a clearer contraction: the probability of collaborating decreases (−0.192), as does internal activity (fewer internal partners: −0.490; lower internal volume: −1.271). For exposed core individuals, the effect mainly appears on the external side, with a decrease in external collaborations (fewer external partners: −0.937; lower external volume: −0.977).

When exposure is measured by intensity, the results become particularly striking for exposed untreated core members: we observe significant and systematic declines, both in individual outcomes (−0.114; −6.071; −0.422) and across all dimensions of collaboration. The probability of collaborating decreases (−0.0065), as do the total number of partners (−0.362) and the total volume (−0.491), with declines on the internal side (internal partners: −0.136; internal volume: −0.275) but also on the external side (external partners: −0.226; external volume: −0.216). For exposed non-core individuals, the most robust effect when exposure is high is a decrease in the volume of internal collaborations (−0.085). *(appendix p.99)*

In sum, this core/non-core heterogeneity indicates that the program mainly benefits central actors when they are direct beneficiaries (a strong increase in collaborations), but that exposure to the treatment can be very unfavorable for untreated core members who are strongly connected to future treated individuals, which is consistent with a dynamic of concentration of collaborations around supported groups and a possible displacement effect for some close but non-beneficiary actors.

Conclusion

In sum, the evaluation suggests that **the reform operates first as a shock to the relational structure rather than as a shock that is immediately visible in individual performance**. The average estimates on individual outcomes do not reveal a clear break after 2012, whereas collaboration indicators react much more clearly: the effect lies mainly on the intensive margin (more partners, more volume), and it is primarily driven by a strong increase in collaborations internal to the treated group, while the external dimension does not increase (and external volume decreases). In other words, **the program is accompanied by an intra-group densification and a reallocation of collaborative effort toward the inside**, rather than by greater openness toward other teams.

This configuration is consistent with several classic mechanisms in networks. An **increase in internal ties amounts to increasing local density and clustering** (more interactions within the same "block"), which can improve coordination, reduce frictions, speed up the circulation of information within the group, and strengthen the capacity to produce collective projects. But this same movement can also reduce the role of "bridge" ties (external connections that serve as shortcuts between communities). When **external ties become rarer** (or lose intensity), one typically observes more informational redundancy locally and less efficient diffusion at the global scale, which can explain (i) the absence of a marked average effect on individual outcomes, if the gains from internal coordination do not translate immediately into measured productivity, and (ii) negative effects for some non-treated individuals. Indeed, in a network, a tie does not benefit only those who carry it: it also generates externalities (of information, reputation, and access to opportunities). If treated members shift their collaborations inward, non-treated partners who were connected to future treated members before 2012 lose part of these externalities and opportunities; this is exactly what the spillover results suggest, especially when exposure is measured in intensity. The heterogeneities support this interpretation: "core" members seem to capture more of the direct gains, which is consistent with a greater ability to leverage opportunities (visibility, access to resources, attractiveness). Conversely, exposure can be unfavorable for some non-beneficiaries who are strongly embedded in environments that are about to be funded, which suggests a reconfiguration of collaborations in favor of supported groups and a polarization between a scientific "core" that benefits more from the funding and a less advantaged periphery.

The identified **limitations point** toward a cautious discussion rather than a challenge to the narrative. First, for some outcomes, the **presence of pre-reform dynamics** weakens a strictly causal reading: part of the results should be interpreted as divergences in trajectories rather than as a break entirely attributable to the program. Second, the **dyadic nature of the data may involve selection issues** (all possible pairs are not necessarily observed), which makes the extensive margin more delicate to interpret. These limitations naturally motivate a change of scale in the analysis. Rather than concluding solely from indicators aggregated at the individual level, Part III returns to the unit of observation closest to the mechanism studied: the dyad, in order to examine directly the formation of ties.

III - Impact of treatment on link formation

We estimate a dyadic difference-in-differences model in which the dependent variable equals one if a collaboration link exists between individuals **i** and **j** in year **t**.

$$y_{ij,g_i,g_j,t} = \alpha + \beta \left|T_{g_i} - T_{g_j}\right| \times Post_t + \lambda(T_{g_i} \times T_{g_j}) \times Post_t + \gamma X_{ij,t} + \delta_t + \theta_{ij} + \varepsilon_{ij,g_i,g_j,t},$$

The treatment structure distinguishes mixed dyads (one treated) and fully treated dyads (both treated), each interacted with the post-2012 indicator, using untreated dyads as the reference category. Dyad fixed effects absorb all time-invariant pair-specific heterogeneity, while year fixed effects control for common macro and network-wide shocks. Identification therefore relies exclusively on within-dyad variation over time, allowing **beta** and **lambda** to capture differential post-policy changes in link formation relative to untreated pairs.

	m_s1	m_s2	m_s12
	Selection 1	Selection 2	Selection 1 & 2
Dependent Var.:	y	y	y
one_treated x post	0.0182* (0.0089)	0.0240. (0.0129)	0.0026 (0.0321)
post x both_treated	0.0391*** (0.0093)	0.0369** (0.0129)	0.0704** (0.0250)
Fixed-Effects:			
id_dyad	Yes	Yes	Yes
year	Yes	Yes	Yes
S.E.: Clustered	by: grou. & grou.	by: grou. & grou.	by: grou. & grou.
Observations	263,634	227,392	71,952
R2	0.16844	0.17919	0.23888
within R2	0.00061	0.00040	0.00150

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

We estimate the model under three alternative sample definitions in order to test the robustness of our results to different selection rules. **Selection 1**. When restricting the sample to dyads satisfying selection_1, we find a positive and statistically significant effect for fully treated dyads (around +3.9 percentage points). Mixed dyads show a smaller and less robust increase. This suggests that the reform primarily strengthens collaboration among pairs where both individuals are treated. **Selection 2**. Using selection_2 leads to very similar conclusions. Fully treated dyads experience a significant post-2012 increase (approximately +3.7 percentage points), while mixed dyads display a modest spillover effect. The similarity across Selection 1 and Selection 2 indicates that the main result is not driven by one-sided inclusion criteria. **Selection 1 & 2** (strict sample). When both selection criteria are imposed simultaneously, the sample becomes smaller but more homogeneous. In this stricter specification, the effect for fully treated dyads increases in magnitude (around +7.0 percentage points) and remains statistically significant, whereas mixed dyads no longer show a significant effect. This indicates that the treatment impact is concentrated primarily among dyads where both members are treated. We therefore choose the **Selection 1 & 2** specification as our preferred model. Although it reduces the number of observations, it ensures a symmetric and economically coherent comparison group, limits potential compositional bias, and provides a more credible "as-if experimental" framework. Importantly, the positive and significant effect for fully treated dyads remains robust; and even stronger; under this stricter definition.

III.1 - Heterogeneity of effects: Within-Group vs Across-Group Dyads

	m_within	m_outside
Dependent Var.:	y	y
post x both_treated	0.0705* (0.0267)	
one_treated x post		-0.0287 (0.0304)
Fixed-Effects:		
id_dyad_in_group	Yes	Yes
year	Yes	Yes
S.E.: Clustered	by: grou. & grou.	by: grou. & grou.
Observations	62,046	11,779
R2	0.24746	0.11353
within R2	0.00110	0.00055

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Investigate whether the policy primarily reinforced internal cohesion or stimulated cross-group integration, we estimate the Difference-in-Differences model separately for within-group and across-group dyads. The results reveal a clear asymmetry. Within groups, the interaction term **post** × **both_treatedis** positive and economically significant (approximately **+7** percentage points), indicating that, after 2012, collaboration probabilities between fully treated individuals belonging to the same group increased substantially relative to untreated dyads. This effect reflects a strong internal reinforcement mechanism, whereby the reform intensified collaboration among members of treated groups. Importantly, mixed dyads cannot exist within groups by construction, so identification in this sample isolates the pure effect of full treatment exposure. In contrast, across-group dyads

display no statistically significant positive effect: the coefficient on **one_treated × post** is negative and insignificant, providing no evidence of outward spillovers from treated to untreated individuals across group boundaries. Taken together, these findings indicate that the reform did not operate through broad network integration, but rather through a consolidation channel, strengthening collaboration within treated groups without generating meaningful cross-group diffusion. This pattern is fully consistent with the structural evidence documented above and reinforces the interpretation of the policy as a catalyst for internal cohesion rather than external expansion.

III.2- Heterogeneity of effects: Core & Gender

a) Heterogeneity by Core

	m_within_corecc	m_within_coremix	m_within_noncc
	Within: core-core	Within: core-noncore	Within: noncore-noncore
Dependent Var.:	y	y	y
post x both_treated	0.2199*** (0.0458)	0.1289*** (0.0359)	0.0223 (0.0271)
Fixed-Effects:			
id_dyad	Yes	Yes	Yes
year	Yes	Yes	Yes
S.E.: Clustered	by: id_1 & id_2	by: id_1 & id_2	by: id_1 & id_2
Observations	14,363	18,267	29,416
R2	0.28608	0.23021	0.24445
Within R2	0.00693	0.00298	0.00014

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Within treated groups, the policy effect is highly concentrated among core actors. The coefficient on **post × both_treatedis** extremely large and precisely estimated for **core–core** dyads (+0.2199, p<0.001), implying that the probability of collaboration between two treated core members increases by approximately 22 percentage points after 2012. This is a quantitatively substantial effect, indicating that the reform strongly reinforced collaboration at the center of treated groups. The effect remains positive and highly significant for **core–noncore** dyads (+0.1289, p<0.001), suggesting that treated core members also intensify collaboration with peripheral members, though to a lesser extent than within the core itself. In contrast, the effect for **noncore–noncore** dyads is small and statistically insignificant (+0.0223), indicating no meaningful internal expansion among purely peripheral actors. Taken together, these results reveal a clear **hierarchical reinforcement** mechanism: the reform strengthens internal collaboration primarily through core actors, who both intensify ties among themselves and **extend connections toward noncore members.** However, peripheral actors do not autonomously increase collaboration absent core involvement. The policy therefore amplifies central actors' coordinating role rather than generating uniform internal densification.

	m_out_corecc	m_out_coremix	m_out_noncc
	Outside: core-core	Outside: core-noncore	Outside: noncore-noncore
Dependent Var.:	y	y	y
one_treated x post	-0.0850 (0.0509)	-0.0236 (0.0554)	0.1404** (0.0508)
post x both_treated		0.0563 (0.0548)	0.1494** (0.0495)
Fixed-Effects:			
id_dyad	Yes	Yes	Yes
year	Yes	Yes	Yes
S.E.: Clustered	by: id_1 & id_2	by: id_1 & id_2	by: id_1 & id_2
Observations	928	5,084	5,767
R2	0.16546	0.09586	0.14334
Within R2	0.00515	0.00395	0.00436

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Across group boundaries, the pattern differs sharply. For **core–core** dyads, the coefficient on one_**treated** × **postis** negative and not statistically significant (-0.0850), providing no evidence that the reform increased cross-group collaboration among core members. If anything, the point estimate suggests a potential contraction, though imprecisely estimated due to limited observations. Similarly, for **core–noncore** dyads across groups, neither treatment interaction is statistically significant, indicating no meaningful spillover-driven expansion of cross-boundary collaboration for mixed-status pairs. Strikingly, the only robust positive cross-group effect appears among **noncore–noncore** dyads. Both interaction terms are positive and statistically significant (≈ +0.14 to +0.15), implying substantial increases in collaboration probabilities between peripheral actors across groups after 2012. This suggests that while core actors consolidate internally, peripheral members may be the primary drivers of cross-group integration.

Combining within- and across-group results reveals a dual mechanism. **Internally**, the reform reinforces collaboration through **core-centered** consolidation, substantially strengthening ties at the top of the network hierarchy. Externally, however, **cross-group expansion** does not originate from core actors but rather from **noncore members**. The treatment therefore produces asymmetric network restructuring: it amplifies centralization and cohesion within treated groups while simultaneously allowing peripheral actors to expand cross-group interactions.

b) Heterogeneity by gender

```
                        m_within_FF      m_within_MM   m_within_mixed
                        Within: F-F      Within: M-M    Within: Mixed
Dependent Var.:                   y                y                y

post x both_treated 0.0728 (0.0461) 0.0700* (0.0290) 0.0638* (0.0322)
Fixed-Effects:      --------------- --------------- ---------------
id_dyad                         Yes             Yes             Yes
year                            Yes             Yes             Yes
_________________ _______________ _______________ _______________
S.E.: Clustered     by: id_1 & id_2 by: id_1 & id_2 by: id_1 & id_2
Observations                  8,030          25,867          26,493
R2                          0.21093         0.26606         0.24083
Within R2                   0.00114         0.00102         0.00100
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Within treated groups, the policy generates a broadly distributed increase in collaboration across gender compositions. The coefficient on **post** × **both_treated is** positive in all three categories—female–female (0.0728), male–male (0.0700*), and mixed-gender (0.0638*) dyads—indicating post-2012 increases of roughly 6–7 percentage points in collaboration probability. The effect is statistically significant for male–male and mixed pairs, while the female–female estimate is of similar magnitude but imprecisely estimated, likely reflecting smaller sample size rather than structural absence of impact. Importantly, the magnitudes are remarkably similar across gender types. This suggests that, **within treated group**s, the reform reinforced collaboration in a relatively gender-neutral manner. The policy does not appear to disproportionately favor a specific gender pairing; rather, it generates a generalized strengthening of internal ties. If anything, the slightly stronger statistical precision among male–male and mixed dyads reflects compositional scale rather than differential treatment intensity. Overall, **within-group** dynamics indicate that the reform operated through **internal consolidation** mechanisms that affected all gender compositions relatively uniformly.

```
                            m_out_FF         m_out_MM     m_out_mixed
                        Outside: F-F     Outside: M-M  Outside: Mixed
Dependent Var.:                   y                y               y

one_treated x post  0.1121** (0.0421)  0.0460 (0.0577) 0.1057 (0.0801)
post x both_treated  0.1050* (0.0509) 0.1195* (0.0570) 0.1191 (0.0791)
Fixed-Effects:      ----------------- --------------- ---------------
id_dyad                           Yes             Yes             Yes
year                              Yes             Yes             Yes
_________________ _________________ _______________ _______________
S.E.: Clustered       by: id_1 & id_2 by: id_1 & id_2 by: id_1 & id_2
Observations                    1,342           5,366           4,854
R2                            0.12783         0.14362         0.09103
Within R2                     0.00169         0.00471         0.00163
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Across group boundaries, the pattern becomes more differentiated. For female–female dyads, the coefficient on **one_treated** × **post** is positive and statistically significant (+0.1121**), implying an increase of roughly 11 percentage points in cross-group collaboration when exactly one member is treated. In addition, **fully treated female dyads** across groups also show a positive and marginally significant increase. This indicates that cross-group expansion among women is particularly responsive to treatment exposure. For **male–male dyads**, the mixed-treatment effect is not significant, but the fully treated effect ($+0.1195^{*}$) is positive and statistically significant, suggesting that cross-group integration among men is concentrated primarily among fully treated pairs rather than through spillovers. **Mixed-gender dyads** display positive coefficients of comparable magnitude (≈10–12 percentage points), yet these effects are not statistically significant, indicating greater variability or weaker identification in this subgroup. Taken together, the cross-group results suggest that spillover effects operate more clearly within **same-gender networks**. Female–female dyads exhibit the strongest mixed-treatment expansion, while male–male dyads respond primarily when both members are treated. Mixed-gender ties show weaker and less precisely estimated responses.

Combining **within- and across-group findings** reveals a nuanced gender mechanism. Internally, the reform strengthens collaboration across all gender compositions without clear favoritism. Externally, however, cross-group integration appears more structured: female networks respond strongly to partial exposure, while male networks expand primarily under full treatment. The **policy** thus reshapes collaboration patterns without inducing gender exclusion, yet spillover channels operate differently across gender-homogeneous versus mixed networks.

Conclusion

To conclude, this report shows that the research funding policy under study manifests itself first as a **change in relational structure** rather than as a break that is immediately visible in individual performance. Over the 2005–2019 period, with 2012 as the **turning point**, the average estimates on individual outcomes do not reveal a clear change after the reform. By contrast, collaboration indicators react much more clearly, which suggests that the reform primarily affects the way researchers organize and connect, and that any translation into measurable individual gains operates through more indirect and potentially slower mechanisms (coordination, complementarities, information circulation, visibility).

This interpretation is reinforced by the very nature of the effects observed on collaborations. The increase is mainly located on the **intensive margin**: after 2012, treated individuals collaborate more in volume and with more partners. Above all, the increase is not symmetric between the inside and the outside of groups: the dynamic is mainly driven by a clear strengthening of **within-group** collaborations within treated groups, while the external dimension does not increase, and external volume decreases. In other words, the intervention is accompanied by a densification "inward" rather than by greater openness toward other teams.

From a network perspective, this pattern is consistent with the idea that an institutional shock can increase **local cohesion**. Strengthening internal ties increases local density and favors more closed structures (more triads), which can improve coordination, reduce frictions, and speed up the circulation of information within groups. Conversely, this reallocation toward the inside can also limit the activation of inter-community connections, which often play an important role in access to distant resources and in the diffusion of ideas at the scale of the whole network. In this setting, it becomes easier to understand why collaborations react strongly while individual outcomes do not show a marked average break: gains in internal coordination do not necessarily translate immediately into measured individual productivity.

The analysis focused on collaborations then confirms that the effect is **concentrated**: the post-2012 increase concerns above all situations where exposure to funding is the strongest, while "mixed" cases display weaker and less robust effects. This conclusion is further refined when distinguishing within-group and across-group ties. The estimates reveal a strong asymmetry: the post-reform increase is substantial **within** treated groups, while there is no convincing evidence of a generalized expansion **across** groups. The policy therefore seems to function as a catalyst for internal cohesion and consolidation, rather than as a lever for global network integration.

Heterogeneities finally provide a more precise reading of the channels. Regarding "core/non-core," it is essential to recall that "core" here refers to **scientific centrality** in the field and in the organization of the group, not to topological centrality in the network. Within treated groups, the effect is massive for core-core dyads and important for core-noncore dyads, while noncore-noncore dyads react little: this suggests a hierarchical

mechanism in which internal intensification is carried by scientifically central actors, who are able to capture and transform opportunities more effectively (projects, visibility, resources). Across groups, when an expansion is detected, it appears to be driven less by core members than by non-core members, indicating that inter-group opening can rely on profiles different from those that structure internal consolidation. Regarding gender, the results indicate a relatively homogeneous internal strengthening across dyad gender compositions, while across-group dynamics are more differentiated, which is consistent with the idea that diffusion channels and spillovers rely on specific relational sub-structures and are harder to identify precisely in some sub-samples.

In this context, spillover results should be read as evidence of **dependence and externalities**, which are not necessarily positive. In a network, a tie does not benefit only those who carry it: it also generates externalities (information, reputation, access to opportunities). If beneficiaries reorient their collaborations inward, some non-treated individuals who were previously connected to future treated individuals may **lose** part of these externalities; the exposure effect is then likely to be **unfavorable** for those most connected to treated actors, which matches the intuition of a concentration of activity around funded groups. In other words, the reform does not only redistribute resources: it also redistributes connections, and therefore opportunities, and this redistribution can create winners as well as losers.

Outlook. It would be particularly interesting to examine more closely how individual productivity evolves as a function of the number of collaborators and the intensity of interactions. Collaboration volume may reflect a trade-off between collective coordination and individual effort: collaborating with more partners can increase complementarities and the diffusion of ideas, but it can also raise coordination costs or dilute each researcher's own effort. Analyzing how individual output varies with the number of co-authors, network position, or scientific centrality would make it possible to deepen our understanding of the underlying microeconomic mechanisms and to better link relational structure and individual performance.